BUSINESS ANALYSIS & SYSTEM DESIGN

Luvuno's Farm Web Ordering System

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Executive Summary

Luvuno’s Farm is a growing agricultural business specializing in poultry, beef, eggs, and pork products. The business currently operates without a digital ordering system, which limits its ability to expand its customer base and compete effectively with established suppliers. Orders are processed manually, and customer engagement depends largely on in-person meetings and phone communication.

This project involved the analysis and design of a web-based ordering system aimed at improving business visibility, enhancing operational efficiency, and providing customers with a seamless purchasing experience. The proposed system allows customers to browse available products, place orders, and make payments online. Additionally, administrative users can manage product listings, monitor orders, and track transactions. The system is intended to support business growth while maintaining operational simplicity.

My Role & Contributions

As the Business Analyst on this project, I was responsible for the end-to-end analysis and system design process. My key contributions included:

- Conducted stakeholder analysis to identify business needs and concerns
- Defined and documented the business problem and opportunity
- Elicited and structured functional and non-functional requirements
- Designed case models and workflow diagrams
- Defined the business rules
- Conducted feasibility and risk analysis
- Designed the system boundary and identified actors
- Defined reporting requirements for the business’s performance

Business Context and Current Challenges

Luvuno currently faces several operational and competitive challenges. The absence of an online presence limits marketing reach and reduces opportunities to attract new customers beyond existing networks. The manual recording of orders increases the likelihood of errors and makes it difficult to track inventory accurately. Furthermore, competing suppliers with established digital platforms can reach customers more efficiently.

The current business process involves scheduling meetings with potential buyers, explaining business operations, negotiating pricing, and manually recording orders.

Payments are verified manually, and delivery scheduling is handled independently by the owner. While this process works on a small scale, it restricts scalability and consumes significant amounts of time and effort.

Project Stakeholders

Stakeholder	Role	Interest & Influence	Stakeholder Concerns	Notes
Business Owner	Manager/ Decision Maker	Wants to grow customer base, increase sales, and enhance operational efficiency	Concerned about revenue, customer acquisition, operational efficiency, and ROI on investments	The owner is not strict on what they are looking for. Interested in the suggestion of the support team.
Customers (Restaurants and Individuals)	Buyers	Require a more convenient route to browse and order meat products	Convenience of ordering, timely delivery, product quality, and pricing	
Investors	Potential future stakeholders	Interested in business growth and online visibility	Business profitability, market visibility, and return on investment	Want to be sure that they are investing in a farm that is confident in bringing back profits.
System Administrator	Managing the website	Tracking orders and updating products	Website downtime, errors in order processing, and product update delays	Want to be involved in every step to understand the system better
Employees	Operations and Delivery	Interested in animal welfare and efficient workflow; influence product quality & customer satisfaction	Animal welfare, safe working conditions, workload, and efficiency of processes	Won't be using the system as much as the administrator
Suppliers	Supply animal feed and other resources	Interested in ongoing business; influencing animal health, growth rate, and product supply	Timely payments, consistent orders, long-term relationships	

Business Opportunity & Problem Statement

Business Opportunity

- Establish an **online presence** to attract more customers
- Expand the **client base** beyond local restaurants

- Capitalize on **emerging markets** such as goat meat exports to the UAE
- Attract **investors** by demonstrating business scalability
- Leverage **sustainable, ethical, or organic farming** as a marketing advantage

Problem Statement
Currently, Luvuno's Farm lacks a website and social media presence, making it difficult to attract new customers and expand the business. The absence of an online ordering system forces customers to rely on manual, inefficient processes.

Summary of Proposed System and Benefits

We propose a website-based system that showcases his products and allows the clients to order products through the website. The admin will also be able to manage the system and product availability. The website allows customers to register, browse products, add selected products to the cart, submit orders, complete payment, and track orders.

Benefits:
- Customers can order at any time, increasing convenience through 24/7-hour access to ordering services
- Improves business visibility and market reach
- The company will be able to engage more with the customers
- The owner will be able to track sales and improve the overall operational efficiency by reducing manual processes.

Overview of Feasibility

The feasibility of this project is evaluated in terms of **technical**, **economic**, and **operational feasibility:**

1. **Technical Feasibility:**
    - The system will be a **web-based** platform, which is achievable with common web technologies (HTML, CSS, JavaScript, Firebase for backend services).
    - The business does not require complex online payment integration, making development simpler.
    - The website will be hosted using affordable hosting solutions to ensure accessibility.
2. **Economic Feasibility:**

- The cost of development is minimal, as the project mainly requires web development efforts and a domain/hosting plan.
- Long-term maintenance costs will be low since product updates can be handled manually by Luvuno or an assigned admin.
- The potential return on investment (ROI) is high as the website will increase visibility and potentially attract more customers.

3. **Operational Feasibility:**
   - The system will streamline the order process, making it easier for customers to purchase products.
   - Luvuno or an employee will be trained to manage the platform, ensuring smooth operation.
   - The system will reduce Luvuno's reliance on manual customer acquisition efforts.

Functional Requirements.

FR1. Product Catalogue Management

The system must display and manage available farm products in a structured and organized manner.

- The system must display products grouped by category (poultry, beef, pork, and eggs).
- The system must display product details, including price, description, and availability status.
- The system must allow administrators to add new products to the system.
- The system must allow administrators to update product prices and stock levels.
- The system must allow administrators to mark products unavailable when out of stock.

FR2. Order Processing

The system must allow customers to select, review, and confirm orders through an online shopping process.

- The system must allow customers to add products to a shopping cart.
- The system must allow customers to modify quantities before checkout.
- The system must automatically calculate the total cost of the order, including VAT and delivery fees.
- The system must generate a unique order number for each confirmed transaction.

- The system must store order details securely in the database.

FR3. Payment Handling

The system must facilitate secure payment options for customers.

- The system must allow customers to select the preferred payment method (EFT, Card, or Cash on Delivery).
- The system must verify payment confirmation before marking an order as paid.
- The system must update the order status once payment has been successfully processed.
- The system must store payment details for administrative tracking purposes.

FR4. Order Tracking & Administration

The system must provide administrative oversight and tracking of customer orders.

- The system must allow administrators to view all orders placed on the system.
- The system must allow administrators to update order status (Pending, Paid, Delivered, or Cancelled).
- The system must allow customers to assess the status of their orders.
- The system must maintain a history of past customer orders for reporting purposes.

Non-Functional Requirements

NFR1. Usability

The client not only sells to restaurant owners, but also to his local community and small business startups.

- The system needs to be easy to use to accommodate the different levels of social stratification within its clientele.
- The font used on the system needs to be readable for all levels of customers.
- Avoid different colour schemes on the page to accommodate users with impaired vision, which is common among our older demographic customers.
- Support multiple language options so a wider demographic can use the website.

NFR2. Reliability

Does the system provide alternative solutions whenever a client encounters an error?

- Once the user has failed to validate their personal details, the system should redirect the user to use a different login method (the use of an email address instead to identify themselves)
- The system must be able to recover from errors without any data loss.

- Ensure you have backup inventory to prevent running short of supplies.

NFR3. Performance

The system's performance is a critical factor in determining its effectiveness and customer satisfaction.

- The results searched must be displayed within 3 seconds for at least 95% of the searches made under normal conditions (4 g mobile network)
- The system should be able to support 50 concurrent users without performance degradation
- The system should be able to ingest data streams from various sources (customer orders, inventory updates, and payment transactions) without loss of data due to connection issues, processing delays, or system crashes.

NFR4. Supportability

For seamless system management, the farmer needs efficient tools to maintain, update, and support the system. A well-designed infrastructure ensures smooth operations and minimizes downtime.

- The system should be able to update product information and add new details.
- The system should have regular backups of important information, allowing for instant recovery of any lost data.
- The system should be ready to support any integration with any 3rd party services (for example, PayPal, Google Analytics/Tableau).

Constraints to the project

1. **Implementation**

- Some rural areas have limited connectivity to the internet, potentially affecting the reach to the client base.
- There are advertising costs involved to reach the audience.
- Many farmers might not have credit cards to purchase on the web

2. **Hardware**

- There's a need to be cognizant of the client base: some farmers use old smartphones with basic features that may not allow for web technologies.
- Integrating multiple features can reduce the system's response rate.
- The system should also be accessible to users regardless of the operating system (OS) that is being used (like Android, Windows, iOS, Linux).

3. Software

- The system should be compatible across different systems, browsers, or device types (like Google Chrome, Microsoft Edge, and Mozilla Firefox). This will help support the user base.
- The system will need to support integration with online payment platforms. (Specific APIs for payment processing services need to be used.)
- The system should implement security measures; the measures will require appropriate security software to protect.

4. Legal Constraints

- The system needs to have accurate products displayed and not be misled according to the Consumer Protection Act 68 of 2008. Also, the company needs to comply with food safety and agricultural regulations.
- The system must always ask for consent from the customers before collecting any personal information and give them access to their information according to the Protection of Personal Information Act (POPIA), Act No. 4 of 2013, and the client data cannot be shared with any 3rd party without consent from the clients.
- The terms and conditions should always be clearly outlined on the website so the customers can read and understand them, because these are the obligations that govern the relationship between the farm and its customers.

Business Rules

- Sales made need to make provision for the increased VAT Rate being 15%
- The admin is the only entity that can upload product details
- Out-of-stock products must be displayed on the website
- The customer will receive an email confirming their payment as well as details of their order.
- Estimated delivery timeline is according to market standards (3-7 business days)
- Orders outside the Mpumalanga are charged an extra R100 for delivery
- All buyers must register with valid user credentials.
- All users can register only 1 account
- Cancellation policies should be clearly defined so customers can be aware of the cancellation conditions.
- If a customer cancels their order within a specific timeframe before delivery and has already made a payment, they are entitled to a refund.

Overview of reports

1. Sales Report

Report Name	Sales Report
Report Description	This Sales Report is used to track the sales performance of the business. Will be able to track: sales performance, revenue generated per product category, and the sales per order type
Report Recipient	Admin
Report Frequency	Monthly
Report Justification	This report tracks the overall financial performance and trends by looking at the sales data. This report helps Luvuno's Farm by ensuring that their sales targets are met and exposes areas of improvement. The report allows the admin to look at the current revenue trends, compare their sales from one period to another, and see if there is a demand for the products being sold. Will be able to determine which order types (bulk/regular orders) generate the most revenue.

Reason for the report	• Provide a sales summary that includes the revenue earned and demand for products (the number of orders that are being processed) • It helps the company see when customers order the most (check out the sales peak), so they can easily manage their time and inventory. • The sales report can also highlight which sales strategies were effective and which were not, enabling the company to refine its approach in the future. • Having the sales data simplifies the process of forecasting future sales and making efficient decisions.

2. Customer Report

Report Name	**Customer Report**
Report Description	Used to illustrate the customers' spending, among our customers, which of our buyers contribute the most to revenue generated.
Report Recipient	Admin
Report Frequency	Monthly
Report Justification	This report provides a comprehensive overview of the company's service performance, focusing on customer satisfaction. The report gives insights into areas like customer satisfaction levels, the company's response time, order frequency, and customer retention rates. By looking at the customer report, Luvuno's Farm will be able to identify key areas of improvement and have a deeper understanding of their customers.
Reason for the report	• Provide an idea of which customer groups account for the most revenue generated. • Address any customer complaints and feedback, and improve the whole customer experience and loyalty. • Identify the frequent customers that contribute significantly to the company.

Risk Analysis

Risk	Threat	Mitigation strategy
Price risk	Medium	Cheap imports that place upward pressure on prices. Lobby with local farmers to decrease prices.
Digital literacy	Low	Have a support page for our older clients to navigate easily on the website
Unauthorized access risk	Medium	Ensure the system requires a strong password combined with alphanumeric and special characters
Stock management risk	Low	Ensure timely updates from the admin by sending out notifications to alert that the stock is running low
User training risk	Low	Have a support page/bot to assist our client in navigating the site
Data Loss	High	Make sure that data is frequently backed up and implement cloud storage to ensure quick recovery in case there are any failures.
Cybersecurity threats	High	Use data encryption and create backups, conduct employee training to enhance awareness, and keep the systems and software updated at all costs
Legal Compliance Risk	Medium	Ensure internal compliance risk policies and procedures are implemented within the organization, and train the staff on the policies and procedures

Scalability Risk	Medium	Create a system that will accommodate business growth and new features.
Reputational Risks	High	Ensure that the company provides high-quality services to keep the customer satisfaction level high. Implement the best security practices to protect the company and customers. Always address customer complaints and feedback.

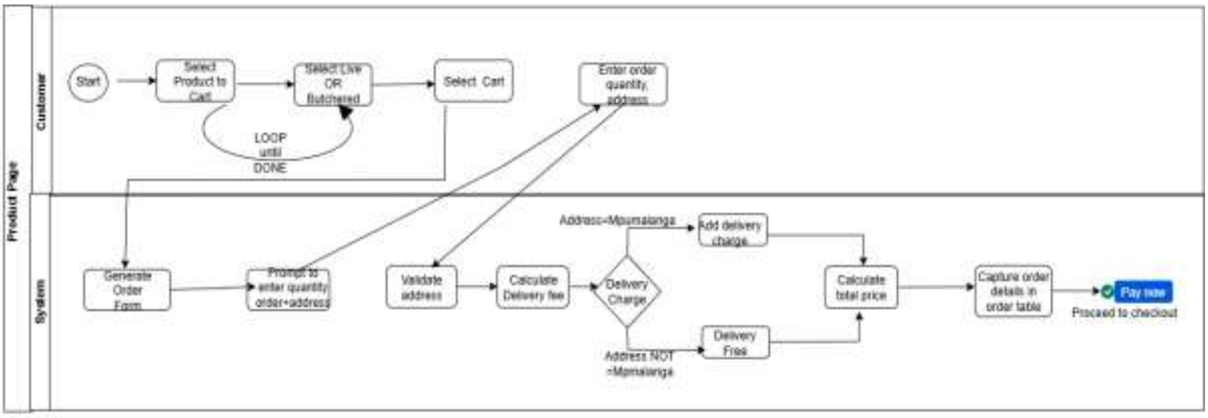
System Boundary

The system that is being developed is a web-based system that will show Luvuno's Farm's products and facilitate orders. The main users of this system are the customers (restaurants, individual buyers), the manager (administrator) who oversees daily operations, and an assistant manager who handles daily operations when the manager is unavailable.

Core Use Case Model

Make Order

Business Process

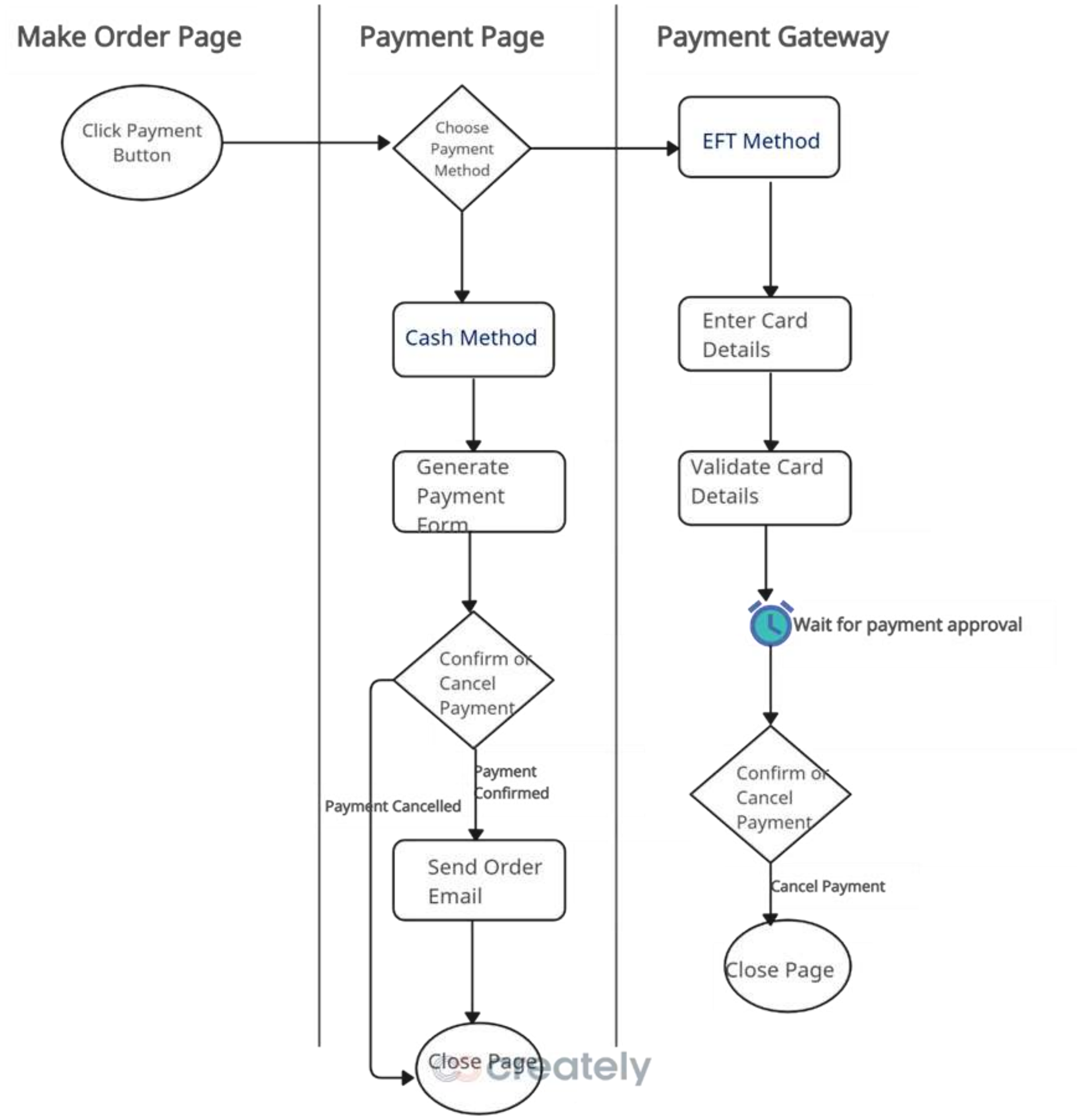


Use Case Name	Make Order
Actors	Customer, System
Trigger	Click ‘Add to Cart’
Brief Description	The customer adds a product(s) to their cart. Once complete they will select the cart icon. The system will generate their order summarry and prompt the user to enter the required order details. The total price will be displayed to the user, in which they will confirm their order. After confirmaton the user will be redirected to the payment page to complete the purchase.

Flow of Activities	Customer	System
	1. Select ‘Add to cart’ on the product page 1.2 Selects their cart 2.1 Select Live or Butchered 2.3 Enter order details: order_quantity, address 2.7. Confirm order 3 Proceed to checkout	1.1. System prompts the user to select their cart 2 Generate order details from the Order Table with the following data fields: order_ID, order_name, product_name 2.2 Prompt user to enter order_quantity and address 2.4 Validate location 2.5 Capture order details into Order Table 2.6 Calculate delivery_price based on location and total_price 2.8 Capture and store delivery_price and total_price into Order table 3.1 Direct customer to payment page
Exceptions	1.2 Select “Add to cart” on product page 2 Enter invalid information	1.2.1 Displays “User not logged in” 1.2.2 Invokes Login use case 2.2 Display system error 2.2.1 Prompt to re-enter information
Pre-conditions	• Customer must exist in the Customer Table • Products must be available in the Product Table	
Post-condition	• order_quantity, address, delivery_price and total_price must be updated in the Order Table	

Make Payment

Business Process

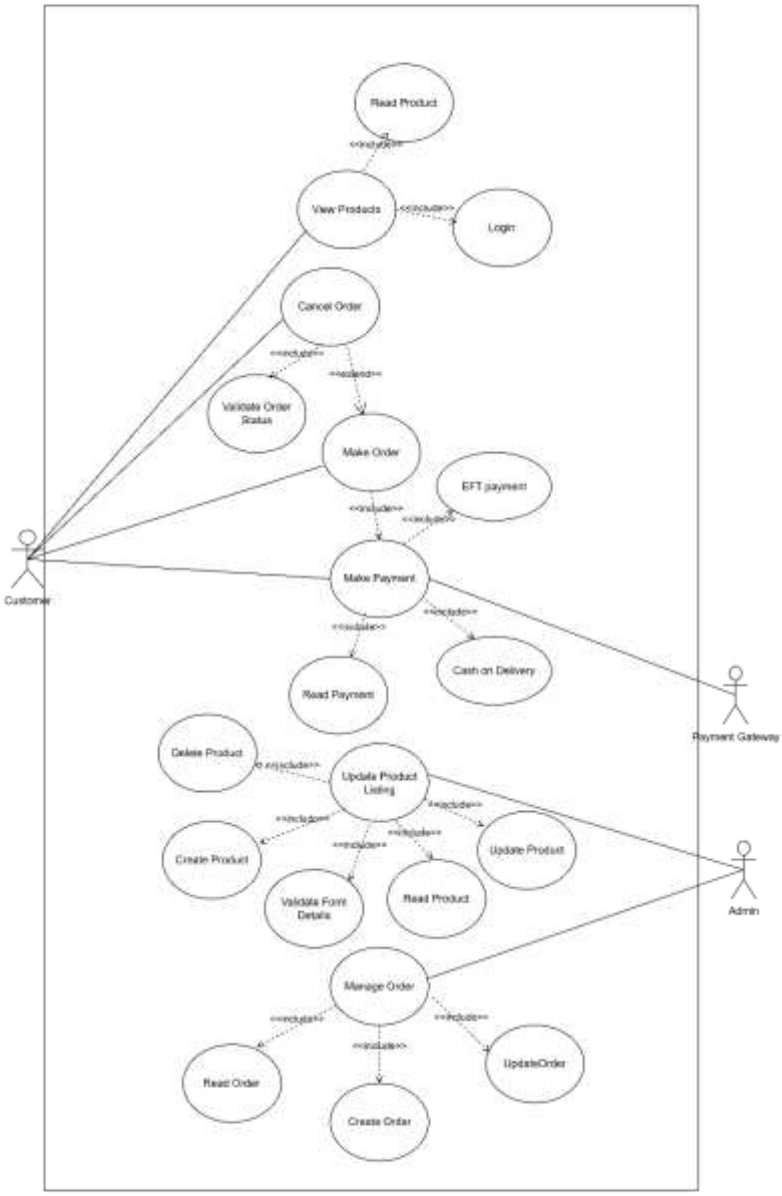


Use Case Name	Make Payment
Actors	Customer, System, Payment Gateway
Trigger	Customer clicks make payment button
Brief Description	The customer clicks the “make a payment” button on the payment page, which redirects the customer to the payment page. They will have the option to either settle their order in cash upon delivery or via eft.

Flow of Activities	Customer	System	Payment Gateway
	1. Customer clicks the “make payment” button	1.1. System redirects the customer to the payment page 1.2. Displays different ways of payment (cash on delivery or EFT)	
	2. [Alt] 2.1. Select “Cash on Delivery” payment method	2.1.1. Generate a form from the Order Table with the following attributes: order_name, order_quantity, order_total, order_type, street_name, Order_suburb, order_city, order_postalCode, payment_type and from the Customer Table with the following attributes: cust_id, cust_name, cust_email 2.1.2. Prompt customer to confirm details 2.1.4. Save Payment option to database 2.1.5. Display confirmation message	
	2.1.3. Customer confirms details	2.1.6. Send confirmation email 2.2.1. Redirect customer to Payment Gateway	

	2.2. Selects EFT payment method 2.2.3. Enter card details 2.2.6. Approve payment	2.2.7. Save payment method to Order Table	2.2.2. Prompt to input card details 2.2.4. Validate card details 2.2.5. Await payment approval
Exceptions	2.1.3. Customer cancels payment 2.2.4. Select ‘Cancel’ button	2.1.4. System closes the payment page 2.2.5. Redirect customer to the payment page	
Pre-conditions	• Order must exist in the Order Table Customer must exist in the Customer Table		
Post-condition	• Confirmation message sent to the customer		

Use Case Diagram

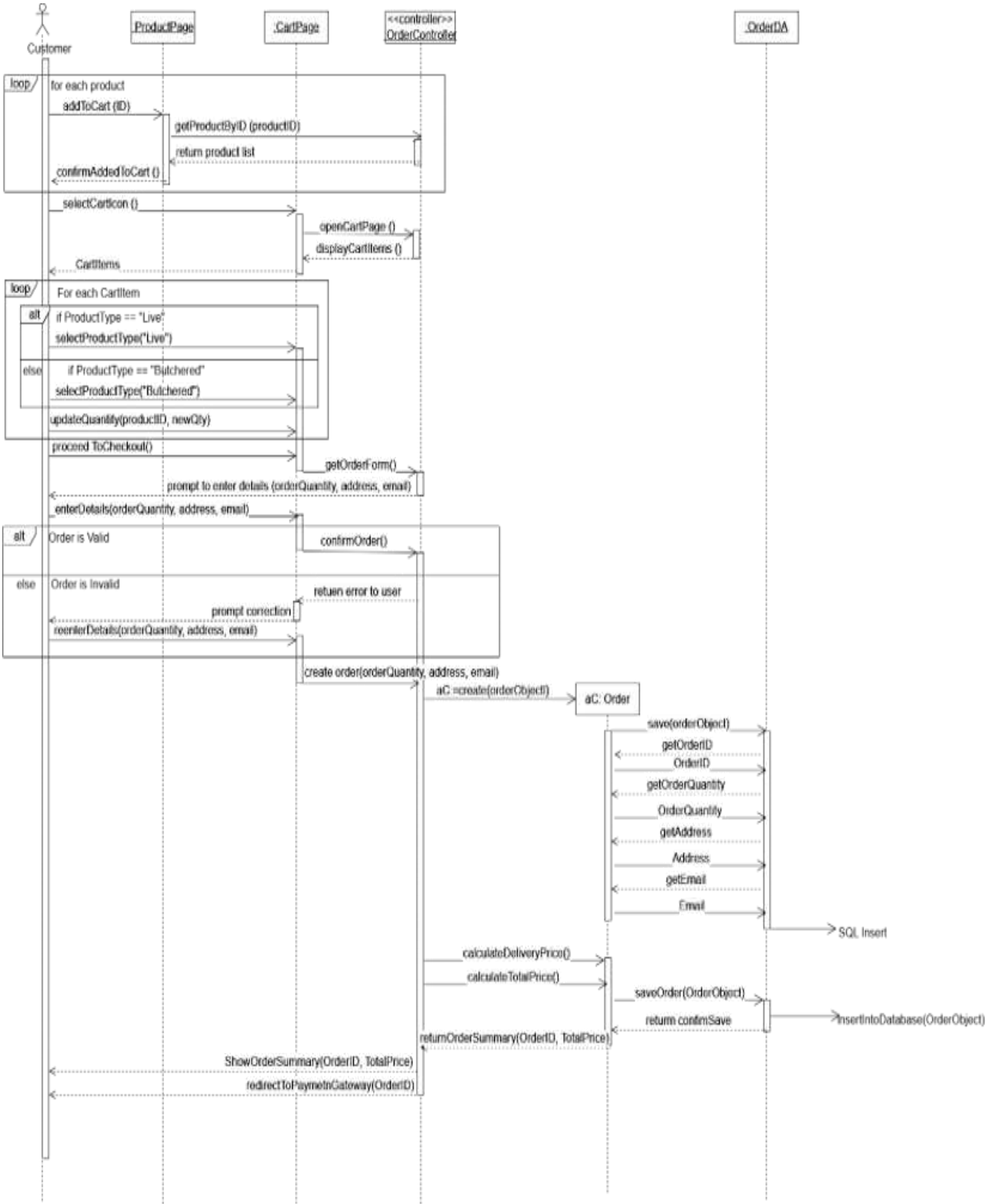


System Feature List

1. **User Authentication:**
    - Customers and administrators will have role-based access.
    - Features include user registration, login, and account management.
2. **Product Catalogue:**
    - The system will display a list of available products, including poultry, eggs, beef, and pork.
    - Products will have detailed descriptions, pricing, and availability.
3. **Order Management:**
    - Customers can add products to their cart and place orders.
    - Admins can view and manage customer orders, including approval or rejection.
4. **Payment Processing:**
    - Customers will use a third-party payment gateway for secure transactions
    - The system will redirect customers to the payment provider
    - The system will store payment confirmation and update order status based on the third party's response
5. **Admin Panel:**
    - Admins will have access to manage product listings, including adding, updating, and deleting products.
6. **Customer Support:**
    - A contact page for inquiries and support requests.
    - Admins can respond to customer inquiries and assist.
7. **Notifications:**
    - The system will send automatic notifications to customers regarding order status, payment verification, and other updates.
8. **Delivery Management:**
    - Admins will manage and schedule product deliveries.
    - A future expansion could involve automated tracking for deliveries.

Sequence Diagrams

Make Order Sequence Diagram



Entity Relationship Data (Data)